QP CODE: 25901396

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth Semester
Value Adddition Course (VAC) - MG4VACMAT200 - Business Mathematics
SCHEME OF VALUATION
Duration: 1 hour 30 minutes Maximum Marks: 50

Part A

Answer any five questions. Each question carries 2 marks.

1. • Frequency Polygon ..... 2 Marks $[K][CO1]$

2. • Class Mark $= \frac{Upper\ limit + Lower\ limit}{2}$. ..... 1 Mark $[U][CO1]$
   • Class Mark of $21 - 25$ is 23. ..... 1 Mark
   • Class Mark of $26 - 30$ is 28. ..... 1 Mark
   (NOTE: Total marks should not exceed 2)

3. Points are $(2, 2)$ and $(3, 7)$ $[U][CO2]$
   • Slope $= \dfrac{y_2 - y_1}{x_2 - x_1}$ ..... 1 Mark
   • $= \dfrac{7-2}{3-2} = 5$ ..... 1 Mark

4. • $y = 2x + 6$ $[U][CO2]$
   • Substituting $x = 1$, $y = 8$, $8 = 2 \times 1 + 6$ ..... 1 Mark
   • $8 = 8$. Hence, satisfied ..... 1 Mark
   • Therefore $(1, 8)$ is a point on the line $y = 2x + 6$ ..... 1 Mark
   (NOTE: Total marks should not exceed 2)

5. • An iso-cost line is a straight line corresponding to an equation obtained by giving a particular value to the objective function Z (cost function) in a Linear Programming Problem. ..... 2 Mark [K][CO3]

   OR

   An iso-cost line is a straight line representing all points which give the same total cost (objective function value) in a Linear Programming Problem. ..... 2 Mark
   OR any other correct statement ..... 2 Mark

6. • Order of $AB = 3 \times 3$. ..... 2 Mark $[K][CO4]$

7. • $|A| = 3 \times 2 - 6 \times 1 = 0$. ..... 1 Mark $[U][CO4]$

   • Hence A is a singular matrix. ..... 1 Mark

8. • The matrix form is $AX = B$ ..... 1 Mark $[U][CO4]$

   • $\begin{bmatrix} -6 & 24 \\ 15 & 5 \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} 213 \\ 525 \end{bmatrix}$ ..... 2 Marks

   OR

   • The matrix form is $AX = B$ ..... 1 Mark

   • $A = \begin{bmatrix} -6 & 24 \\ 15 & 5 \end{bmatrix}, X = \begin{bmatrix} p \\ q \end{bmatrix}, B = \begin{bmatrix} 213 \\ 525 \end{bmatrix}$ ..... 2 Marks

   (NOTE: Total marks should not exceed 2)

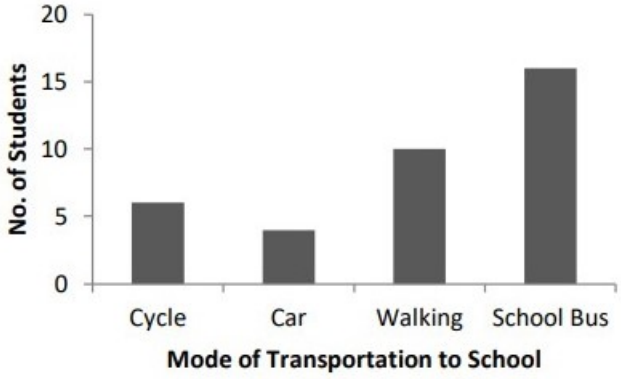
Part B

Answer any four questions. Each question carries 5 marks.

9. • Marking items on X-axis and values on Y-axis. ..... 1 Mark $[U][CO1]$

   • Drawing bars of correct height for each item ..... $1 \times 4 = 4$ Marks

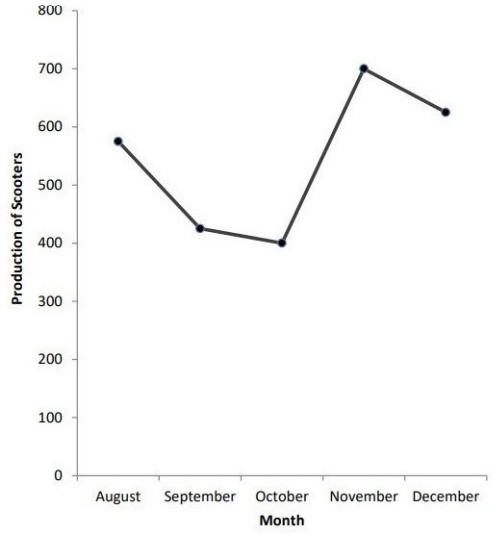
   • Shading all the bars in the same manner ..... 1 Mark



(NOTE: Total marks should not exceed 5. Also, marks should be given for correct figures using any other scale for the values.)

10. • Marking months on X-axis and values on Y-axis. ..... 1 Mark $[U][CO1]$

    • Marking correct points and drawing line segments joining the points ..... 4 Marks



(NOTE: Marks should be given for correct figures using any other scale for the values)

11.
- Finding $x = 7$ ..... 3 Marks [U][CO2]
- Finding $y = 5$ ..... 2 Marks

OR

- Adding the two equations and obtaining $2x = 14$ ..... 2 Marks
- Finding $x = 7$ ..... 1 Mark
- Substituting $x = 7$ in any of the two equations and getting $y = 5$ ..... 2 Marks

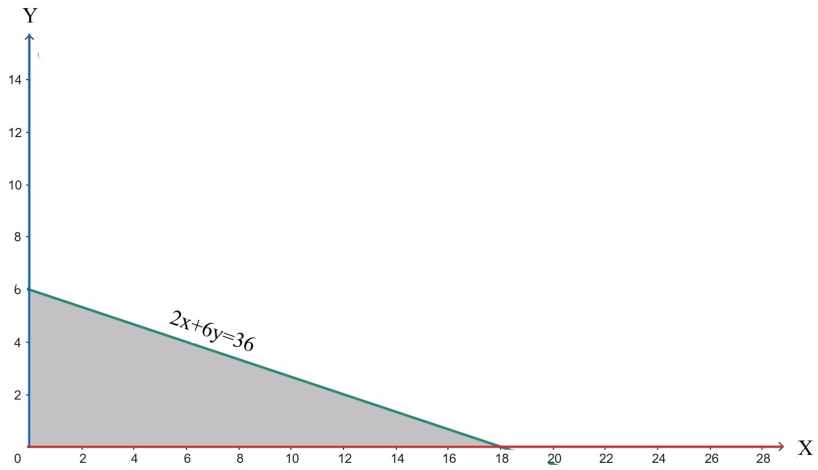
OR

- Writing the matrix form $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 12 \\ 2 \end{bmatrix}$ ..... 1 Marks
- Finding $|A| = -2$ ..... 1 Mark
- Finding the cofactor matrix $C = \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$ ..... 1 Mark
- Finding $A^{-1} = \dfrac{adj(A)}{|A|} = \dfrac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$ ..... 1 Marks
- $X = A^{-1}B = \dfrac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 12 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$ ..... 1 Mark

(NOTE: Appropriate stage marks should be given for applying Cramer's rule or any other correct method)

12.
- Budget constraint is $2x + 6y \leq 36$ ..... 1 Mark [U][CO3]
- Budget line is $2x + 6y = 36$ ..... 1 Mark

- Finding points on the line $2x + 6y = 36$ ..... 1 Mark
- Drawing the budget line ..... 2 Marks
- Shading the budget set ..... 1 Marks



(NOTE: Total marks should not exceed 5)

13. • $2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \\ -4 & 4 \end{bmatrix}$ ..... 2 Marks [U][CO4]

• $2A - B = \begin{bmatrix} 0 & 3 \\ 6 & 5 \\ -9 & -3 \end{bmatrix}$ ..... 3 Marks

14. • Changing the two equations to $6x_1 + 12x_2 = 30$ and $6x_1 + 10x_2 = 24$ ..... 1 Mark [U][CO4]

• Subtracting and getting, $x_2 = 3$ ..... 2 Marks

• Substituting $x_2 = 3$ in any of the two equations and getting $x_1 = -1$ ..... 2 Marks

OR

• Writing the matrix form $\begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 12 \end{bmatrix}$ ..... 1 Marks

• Finding $|A| = -2$ ..... 1 Mark

• Finding the cofactor matrix $C = \begin{bmatrix} 5 & -3 \\ -4 & 2 \end{bmatrix}$ ..... 1 Mark

• Finding $A^{-1} = \dfrac{adj(A)}{|A|} = \dfrac{1}{-2} \begin{bmatrix} 5 & -4 \\ -3 & 2 \end{bmatrix}$ ..... 1 Marks

- $X = A^{-1}B = \frac{1}{-2}\begin{bmatrix} 5 & -4 \\ -3 & 2 \end{bmatrix}\begin{bmatrix} 10 \\ 12 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ .....1 Marks

(NOTE: Appropriate stage marks should be given for applying Cramer's rule or any other correct method)

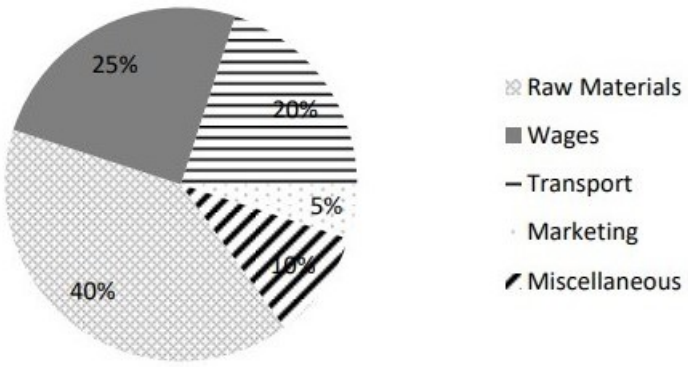
Part C

Answer any two questions. Each question carries 10 marks.

15. • Finding the central angles .....5 Marks [U][CO1]

Item	Percentage	Central Angles
Raw Materials	40%	144°
Wages	25 %	90°
Transport	20 %	72°
Marketing	5 %	18°
Miscellaneous	10 %	36°

- Correct Figure and shading .....5 Marks



16. • Let x_1 and x_2 represent the number of units of product A and B to be produced respectively. ..... 1 Mark [A][CO3]

The LPP is

- Maximize $Z = 7x_1 + 21x_2$ .....2 Marks

subject to

- $5x_1 + 2x_2 \leq 40$ .....2 Marks
- $3x_1 + 12x_2 \leq 60$ .....2 Marks
- $4x_1 + 8x_2 \leq 48$ .....2 Marks
- $x_1 \geq 0,\ x_2 \geq 0$ .....1 Mark

17.
- $|A| = -161$ ..... 2 Marks [U][CO4]
- Cofactor Matrix, $C = \begin{bmatrix} 7 & -49 & 21 \\ -31 & 10 & -1 \\ 11 & 38 & -36 \end{bmatrix}$ ..... 4 Marks
- Adjoint Matrix, $adj(A) = C^T$ ..... 1 Mark
- $= \begin{bmatrix} 7 & -31 & 11 \\ -49 & 10 & 38 \\ 21 & -1 & -36 \end{bmatrix}$ 1Marks
- Inverse Matrix $A^{-1} = \dfrac{adj(A)}{|A|}$ ..... 1 Mark
- $= \dfrac{-1}{161} \begin{bmatrix} 7 & -31 & 11 \\ -49 & 10 & 38 \\ 21 & -1 & -36 \end{bmatrix}$ ..... 1 Marks

18.
- $A^T = \begin{bmatrix} 14 & 6 & 20 \\ 8 & 19 & -1 \\ -3 & 11 & 18 \end{bmatrix}$ ..... 2 Marks [U][CO4]
- $(A^T)^T = \begin{bmatrix} 14 & 8 & -3 \\ 6 & 19 & 11 \\ 20 & -1 & 18 \end{bmatrix}$ ..... 2 Marks
- Thus $(A^T)^T = A$. Verified ..... 1 Mark
- $B^T = \begin{bmatrix} 9 & -8 & 13 \\ 4 & 16 & 2 \\ 5 & 7 & 12 \end{bmatrix}$ ..... 2 Marks
- $(B^T)^T = \begin{bmatrix} 9 & 4 & 5 \\ -8 & 16 & 7 \\ 13 & 2 & 12 \end{bmatrix}$ ..... 2 Marks
- Thus $(B^T)^T = B$. Verified ..... 1 Mark

NB: Suitable Stage Marks maybe fixed by the examiner for alternate correct answers